

7.12 class - 3

Problem: Use the method of Frobenius to find solution of the DE, $2x^2 y'' - xy' + (x-5)y = 0$ in some interval $0 < x < \infty$

Soln:

Given DE, $2x^2 y'' - xy' + (x-5)y = 0$ — (1)

$$\Rightarrow y'' - \frac{1}{2x} y' + \frac{x-5}{2x^2} y = 0$$

Since, $x=0$ is a regular singular point of the given

DE (1), let the solution of the form of Eq (1) is

$$y = \sum_{n=0}^{\infty} C_n x^{n+r} \quad \text{--- (2) then from Eq (1) } \Rightarrow$$

$$y' = \sum_{n=0}^{\infty} C_n (n+r) x^{n+r-1}$$

$$y'' = \sum_{n=0}^{\infty} C_n (n+r)(n+r-1) x^{n+r-2}$$

$$2 \sum_{n=2}^{\infty} C_n (n+r)(n+r-1) x^{n+r} - \sum_{n=1}^{\infty} C_n (n+r) x^{n+r}$$

$$+ 5 \sum_{n=0}^{\infty} C_n x^{n+r}$$

$$\Rightarrow \sum_{n=0}^{\infty} \left[2(n+\pi)(n+\pi-1) - (n+\pi) - 5 \right] C_n x^{n+\pi} + \sum_{n=1}^{\infty} C_{n-1} x^{n+\pi} = 0$$

• $\pi = 0 :$

$$\left[2\pi(\pi-1) - \pi - 5 \right] C_0 x^{\pi} + \sum_{n=1}^{\infty} \left[2(n+\pi)(n+\pi-1) - (n+\pi) - 5 \right]$$

$$C_n x^{n+\pi} + \sum_{n=1}^{\infty} C_{n-1} x^{n+\pi} = 0$$

$$\Rightarrow \left[2\pi(\pi-1) - \pi - 5 \right] C_0 x^{\pi} + \sum_{n=1}^{\infty} \left[\left\{ 2(n+\pi)(n+\pi-1) - (n+\pi) - 5 \right\} C_n + C_{n-1} \right] x^{n+\pi} = 0$$

Equating to zero the coefficient of the lowest power of x , we get indicial equation

$$2\pi(\pi-1) - \pi - 5 = 0 \quad \sum_{n=0}^{\infty} = 0$$

$$\Rightarrow 2\pi^2 - 3\pi - 5 = 0 \quad \sum_{n=0}^{\infty} = 0$$

$$\pi_1 = 5/2 \quad \text{and} \quad \pi_2 = -1/2$$

Also, equating to zero the coefficient of the highest power of x , we get recurrence formula

$$\left[2(n+r)(n+r-1) - (n+r) - 5 \right] C_n + C_{n-1} = 0, \quad n \geq 1 \quad \text{--- (3)}$$

Letting $r = r_1 = 5/2$ in Eq (3) $\Rightarrow$

$$n(2n+7) C_n + C_{n-1} = 0$$

$$C_n = - \frac{C_{n-1}}{n(2n+7)}, \quad n \geq 1$$

Putting $n = 1, 2, 3, \dots$, $C_1 = -\frac{C_0}{3}$, $C_2 = -\frac{C_1}{22}$

$$= \frac{C_0}{198}$$

$$C_3 = -\frac{C_2}{37} = -\frac{C_0}{7722}$$

Now letting $r = 5/2$ and using these values of

$C_1, C_2, C_3, \dots$ we obtain the solution from (2)

$$y = C_0 \left(x^{5/2} - \frac{1}{3} x^{7/2} + \frac{1}{198} x^{9/2} - \dots \right) = C_0 x^{5/2} \left(1 - \frac{1}{3} x + \frac{1}{198} x^2 - \dots \right)$$

we now, let

$$r = r_2 = -1 \text{ in Eq (3)}$$

$$n(2n-7)c_n + c_{n-1} = 0, \quad n \geq 1$$

$$c_n = - \frac{c_{n-1}}{n(2n-7)}, \quad n \geq 1$$

$$\text{Putting } n=1, 2, 3 \dots c_1 = \frac{1}{3}c_0, c_2 = \frac{1}{6}c_1 = \frac{1}{30}c_0$$

$$\Rightarrow c_3 = \frac{1}{3}c_2 = \frac{1}{90}c_0, \dots$$

letting $r = -1$ in (2) and using these ~~the~~ values of $c_1, c_2, c_3, \dots$ we get another solution of the form,

$$y = y_2 = c_0 \left(x^{-1} + \frac{1}{3}x + \frac{1}{30}x^3 + \frac{1}{90}x^5 \dots \right)$$

$$y = y_2 = c_0 x^{-1} \left(1 + \frac{1}{3}x + \frac{1}{30}x^3 + \frac{1}{90}x^5 \dots \right)$$

Thus the general solution of Eq (1) is

$$y = c_1 y_1(x) + c_2 y_2(x) \text{ (General Solution)}$$

$\hookrightarrow y_2$ and y_1 are linearly independent

Note :

Case - I

Suppose, $\pi_1 - \pi_2 \neq N$, where N is non-negative integer, then the DE has two linearly independent solutions

$$y_1 \text{ \& } y_2 \text{ of the form } y_1(x) = \sum_{n=0}^{\infty} c_n (x-x_0)^n ; c_0 \neq 0$$

$$\text{and } y_2(x) = \sum_{n=0}^{\infty} c_n^* (x-x_0)^{n+\pi_2} ; c_0 \neq 0$$

Case - II

Suppose, $\pi_1 - \pi_2 = N$, where N is a (+)ve integer, then the DE has two linearly independent solutions of the form,

$$y_1(x) = \sum_{n=0}^{\infty} c_n (x-x_0)^{n+\pi_1}, \quad c_0 \neq 0 \text{ and}$$

$$y_2(x) = \sum_{n=0}^{\infty} c_n^* (x-x_0)^{n+\pi_2} + c_y \ln |x-x_0|$$

Case - II

Suppose $\pi_1 - \pi_2 = 0$ then the given DE has two linearly independent solution:

$$y_1(x) = \sum_{n=0}^{\infty} C_n (x-x_0)^{n+\pi_1} \quad C_0 \neq 0$$

$$y_2(x) = \sum_{n=0}^{\infty} C_n^* (x-x_0)^{n+\pi_2-1} + y_1(x) / n(x-x_0)$$